Thromboprophylaxis: Non-Orthopedic Surgery

Objective:

To outline a practical approach for the prevention of venous thromboembolism (VTE) in patients undergoing non-orthopedic surgery.

Background:

VTE is a common and yet generally preventable cause of post-operative morbidity and mortality.

The use of graduated compression stockings (GCS), intermittent pneumatic compression (IPC), low-dose unfractionated heparin (UFH), and low molecular weight heparin (LMWH), combined with early ambulation, have all been shown to reduce post-operative VTE to varying degrees in patients undergoing surgery.

General Principles of Prophylaxis:

1. Thromboprophylaxis decisions rely on a consideration of the balance in the risks of VTE and bleeding, both of which are affected by procedure-specific and patient-specific factors.
2. Patients at moderate-to-high VTE risk without a high bleeding risk should receive pharmacological (anticoagulant) thromboprophylaxis:
   - Suggest using LMWH (preferred option in most patients) or UFH (see anticoagulant dosing on page 4)
   - The use of direct oral anticoagulants (e.g. apixaban, dabigatran, edoxaban, rivaroxaban) has not been well evaluated for thromboprophylaxis in the non-orthopedic surgical setting for thromboprophylaxis
3. Patients at moderate-to-high VTE risk with high bleeding risk features (see 4, below) should receive mechanical thromboprophylaxis:
   - Intermittent Pneumatic Compression (IPC) devices OR
   - Properly fitted bilateral, calf-length Graduated Compression Stockings (GCS), if IPC devices are not available

     Mechanical thromboprophylaxis should be used continuously except for bathing and walking (for IPC devices).
4. Strong contraindications to pharmacologic thromboprophylaxis (high risk bleeding features) include clinically important bleeding, severe thrombocytopenia (25 x 10^9), and untreated major bleeding disorders. Relative contraindications to anticoagulant thromboprophylaxis include recent intracranial hemorrhage, recent paraspinal bleeding, and recent high bleeding risk surgery. Mechanical thromboprophylaxis is contraindicated in patients with lower limb arterial insufficiency or peripheral neuropathy or when deep vein thrombosis is suspected.
5. For most elective non-orthopedic surgery patients in whom pharmacological thromboprophylaxis is recommended, thromboprophylaxis should be initiated when hemostasis is achieved (for patients without a high risk of bleeding consider starting within 12 hours of surgery). In patients in whom mechanical thromboprophylaxis is recommended, start at the beginning of the procedure. For patients admitted to hospital before surgery, assess for pre-operative pharmacological thromboprophylaxis (see **Clinical Guide Thromboprophylaxis: Hospitalized Medical Patients**).

6. Although the optimal duration of thromboprophylaxis is not well defined, patients with a mo
for VTE should receive thromboprophylaxis at least until discharge from hospital. Extendec
thromboprophylaxis (up to 30 days) may be considered in select populations.

Thromboprophylaxis Approaches in Non-Orthopedic Surgery:

There are three general approaches to thromboprophylaxis in patients undergoing non-orthopedic surgery.

1. **Group-based:** This approach provides standard thromboprophylaxis to all patients who "belong to" a large group (e.g. abdominal-pelvic surgery, spine surgery, etc.) unless there is a specific patient contraindication. This approach has been adopted by many hospitals in developing their own prophylaxis guidelines.
2. **Individualized:** This strategy is based on individual patient risk assessment using a formal risk assessment model like the Caprini score ( ). This score has been validated for general, urologic, and gynecologic surgeries. In general, the estimated VTE risk and thromboprophylaxis recommendations using this approach are similar to the group-based approach.
3. **The American Society of Hematology (ASH) 2019 guidelines** for management of VTE use a blended approach that incorporates both group-based and individualized recommendations, as summarized in **Table 1** below.

Risk of bleeding assessments have not been validated in a formal model for surgical patients.

It is suggested that every institution have a written policy for thromboprophylaxis and, where possible, that thromboprophylaxis protocols be embedded into routinely used electronic or paper order sets.

Evidence is lacking as to whether thromboprophylaxis is needed in neonates and children who have non-orthopedic surgery. However, there may be high-risk cohorts in whom thromboprophylaxis may be considered. Consultation with a pediatrician or hematologist with expertise in pediatric thrombosis is recommended.

Table 1: Surgical site-specific recommendations:

*As noted in General Principles, when pharmacological thromboprophylaxis is recommended, LMWH or UFH is suggested over the use of DOACs

**In patients not receiving pharmacological thromboprophylaxis, mechanical thromboprophylaxis should be used initially. Bleeding risk should be reassessed frequently with consideration of pharmacological thromboprophylaxis at that time if appropriate.

Thromboprophylaxis in bariatric surgery has not been well studied. Based on retrospective data, we suggest thromboprophylaxis for 10 days with:

Tinzaparin 75 U/kg SC once daily or 4,500 U SC once daily for patients <110 kg
10,000 U SC once daily for patients 110-159 kg
14,000 U SC once daily for patients ≥160 kg

Anticoagulant Dosing:

LMWH usual doses:

- dalteparin 5,000 U SC once daily
- enoxaparin 40 mg SC once daily
- tinzaparin 4,500 U SC once daily

LMWH for weight <40 kg (reduce to next lower pre-filled syringe dose):

- dalteparin 2,500 U SC once daily
- enoxaparin 30 mg SC once daily
- tinzaparin 2,500 U SC once daily (~75 U/kg)

LMWH for weight 100 - 120 kg:

- dalteparin 7,500 U SC once daily
- enoxaparin 60 mg SC once daily
- tinzaparin 8,000 U SC once daily

LMWH for weight >120 kg:

- dalteparin 7500 U SC daily (there is no data for higher doses up to 180kg)
- enoxaparin 0.5mg/kg SC once daily to BID (depending on thrombotic risk)
- tinzaparin 75 U/kg SC daily

LMWH with severe renal dysfunction (CrCl <30 mL/min):

- no dosage reduction for dalteparin or tinzaparin (preferred option)
- enoxaparin 30 mg SC once daily

Unfractionated heparin is usually given as 5,000 U SC BID or TID (in higher risk patients).

Consider higher dose (e.g. 7,500 U SC BID or TID, depending on weight and thrombotic risk) for patients with weights >100 kg and lower dose (e.g. 2,500 U SC BID) in patients <40 kg.

Heparin-induced thrombocytopenia (HIT):

- Fondaparinux is used as an alternative to UFH or LMWH for thromboprophylaxis in patient with a history of HIT (usual dose of 2.5 mg SC daily).

Other Relevant

- Thromboprophylaxis: Hospitalized Medical Patients
- Thromboprophylaxis: Orthopedic Surgery
- Unfractionated Heparin and Low-Molecular-Weight Heparin

Relevant Clinical Tool:

- Order Set: Extended Thromboprophylaxis for Patients after Abdomino-Pelvic Cancer Surgery

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